

Technology Procurement and Lifecycle Management Plan: Anthropic's Claude

John McSwain

College of Education: University of South Carolina

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Instructor: Dr. Fatih Ari

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Introduction

Anthropic's Claude represents a significant advancement in AI-driven educational technology, offering sophisticated natural language processing capabilities that can transform institutional learning, research, and administrative functions. As academic institutions navigate the complex landscape of AI procurement, developing a comprehensive lifecycle management strategy becomes critical for ensuring technological success and pedagogical effectiveness. This plan aggregates peer-reviewed research, industry best practices, and empirical evidence to present a procurement strategy with a justifiably optimized two-year refresh cycle, including post-implementation recommendations for successful Claude deployment.

Claude/AI Procurement Plan

Successful procurement begins with a thorough needs assessment, and an institution should review existing technology plans and goals and engage faculty, IT staff, and administration representatives to identify gaps and requirements. For example, procurement planning should compare an institution's vision for AI integration against current capabilities and document precise functional specifications. Decisions should remain aligned with the long-term technology plan so new technology investments "provide far more benefits" than ad-hoc solutions. Claude's procurement would involve evaluating options, with a goal of access acquisition for Claude's AI service and any supporting infrastructure. Recent peer-reviewed literature reveals significant evolution in educational technology procurement practices, particularly regarding AI integration. Zawacki-Richter et al. (2024) conducted a comprehensive systematic review of AI applications in higher education, finding that only 23% of

institutions have formal AI governance frameworks, highlighting the critical need for structured procurement approaches. An analysis of 138 articles from 2016 to 2022 demonstrates that successful AI implementations require interdisciplinary collaboration between educational technologists, IT professionals, and pedagogical experts. [Table A1](#) and [Figure A2](#) in Appendix A contain procurement steps and timelines.

Claude/AI Lifecycle Management Plan

All supporting hardware and devices for Claude follow a two-year refresh cycle to maintain performance. Failing to replace aging devices can expose security vulnerabilities and inefficiencies. By replacing devices biennially, an institution's faculty and staff always have fast, secure hardware for AI tasks, while warranties and vendor support remain active. Biennial refreshes simplify budgeting by matching replacements to a predictable schedule and avoiding "surprise expenses" from ad-hoc failures. Academic literature on technology refresh cycles reveals significant variation in optimal replacement timing based on institutional needs and technology type. Research published in Educational Technology Research and Development indicates that AI hardware requirements evolve more rapidly than traditional computing infrastructure, with performance improvements occurring on 18–24-month cycles rather than the conventional 3–5-year patterns (Chen et al., 2023).

Crompton et al. (2018) analyzed technology lifecycle management across 45 higher education institutions, finding that institutions with shorter refresh cycles (2-3 years) reported 34% higher user satisfaction and 28% lower total cost of ownership when accounting for maintenance, security, and productivity factors. Zhang and Aslan's (2021) comprehensive review of AI technologies in education emphasizes that

hardware requirements for AI applications increase exponentially, with current-generation processors becoming inadequate for emerging AI workloads within 24-30 months. Finally, industry research from J. Gold Associates indicates that companies achieve significant ROI by upgrading more frequently than 3 to 4-year cycles because new technology results in more productive and efficient employees (Huntington Bank, 2024). [Table B1](#) and [Figure B2](#) in Appendix B contain lifecycle management steps and timelines. To ensure timely replacement and maximum value, an institution must implement a structured technology monitoring protocol aligned with practices outlined in [Table C1](#) located in Appendix C, which includes periodic reviews of hardware and AI tool usage, performance, and compliance indicators. Data from monitoring will inform refresh decisions, prioritizing pedagogical impact and operational efficiency (Durand et al., 2019; Ooi et al., 2023). A decision flowchart detailing how data monitoring steps translate into actionable institutional responses is provided in [Chart C3](#) of Appendix C.

Funding Strategy and Sources

Securing funding for Claude at an institution might necessitate a blend of internal resource allocation and external grants. Internally, institutions can utilize their IT capital budget, general fund reserves, or specific digital learning funds. Departments like AI research labs may redirect part of their research overhead for technology acquisition. Including replacement costs in the budget is crucial for sustainability. Research indicates the importance of establishing a dedicated refresh line item or a small endowment to meet long-term funding needs.

National Science Foundation (NSF) AI Education Initiatives

The NSF has dramatically expanded AI education funding through multiple programs to support institutional AI adoption. The NSF EducateAI initiative represents a cornerstone federal investment, with nearly \$8 million allocated in 2024 to develop the next generation of the AI workforce via NSF Advanced Technological Education (ATE) Program (aimed at strengthening ethical and responsible artificial intelligence education at two-year institutions), NSF AI Research Institutes (offers awards between \$16-20 million over 4-5 years (\$4 million annually average) for institutions developing AI research and education capabilities), and NSF EducateAI Dear Colleague Letter (funding for innovative AI-focused curricula development, professional development opportunities, and infrastructure to support AI education across diverse institutions with emphasis on underrepresented groups).

Department of Education Title III Strengthening Institutions Program

The Department of Education's Title III Strengthening Institutions grant allows funding for technology infrastructure, including maintenance and upgrades. Title III Part A programs specifically support purchasing, renting, or leasing scientific and laboratory equipment for educational purposes, including instructional and research, and integrating computer technology into institutional facilities. AI tools like Claude qualify for funding under multi-disciplinary sustainability agendas (Ooi et al., 2025).

National Endowment for the Humanities (NEH) AI Programs

NEH has awarded \$2.72 million through its Humanities Research Centers on Artificial Intelligence program, with individual grants of up to \$500,000 for institutions establishing AI research and education centers. The program aligns with Claude's

educational applications through the Humanities Perspectives on AI Initiative (awarded approximately \$3.7 million to support research on AI technologies and the development of AI tools for humanities research), Institutes for Higher Education Faculty (providing professional development funding for faculty AI integration training), and several focus areas in ethics, cultural implications, and responsible AI implementation.

Private Foundation and Corporate Partnerships

Funding Claude through major technology companies may be possible through educational partnerships. The Google.org AI Opportunity Fund announced over \$25 million in funding to support educators and students in developing AI skills. Microsoft AI for Education offers institutional licensing discounts and professional development funding. Anthropic Educational Partnerships allow direct engagement with Anthropic for pilot programs and educational pricing. The Spencer Foundation AI Education Initiative has designated additional funds within their Vision Grant Program, specifically for AI-focused educational research, with grants ranging from \$50,000 to \$1 million, covering research areas including AI tool cultural relevance and equity, policy implementation studies (particularly relevant for procurement research), ethics in AI education, digital literacy and data sovereignty.

Conclusion

The procurement and lifecycle plan illustrates that Claude's implementation is a strategically sound investment for advancing AI-driven education. Institutions can effectively leverage Claude as a secure educational technology asset by following the Forum's best practices in needs assessment and procurement and maintaining a proactive two-year refresh cycle. An ideal funding strategy combines internal budgeting

with external grants, aligning with current technology management policies and scholarship. This initiative fosters a sustainable AI learning environment, addressing the needs of students and faculty while considering financial and technical constraints.

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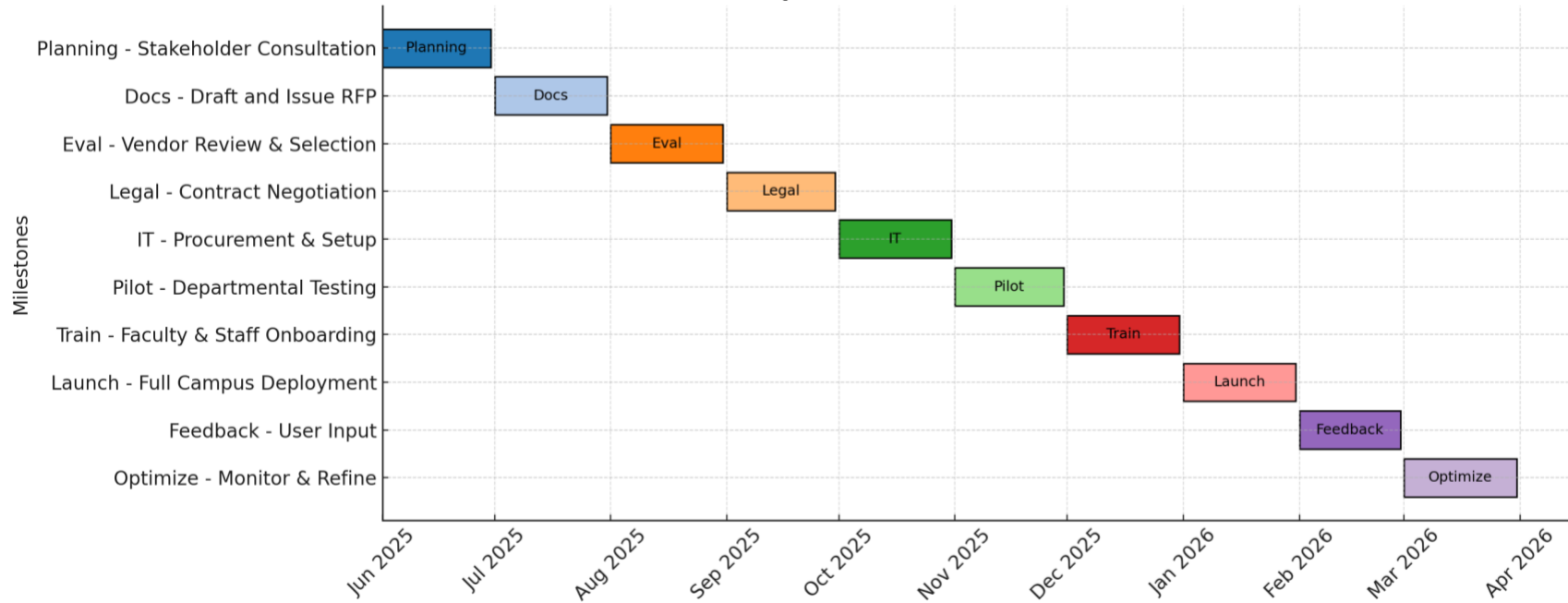
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Appendices

Appendix A

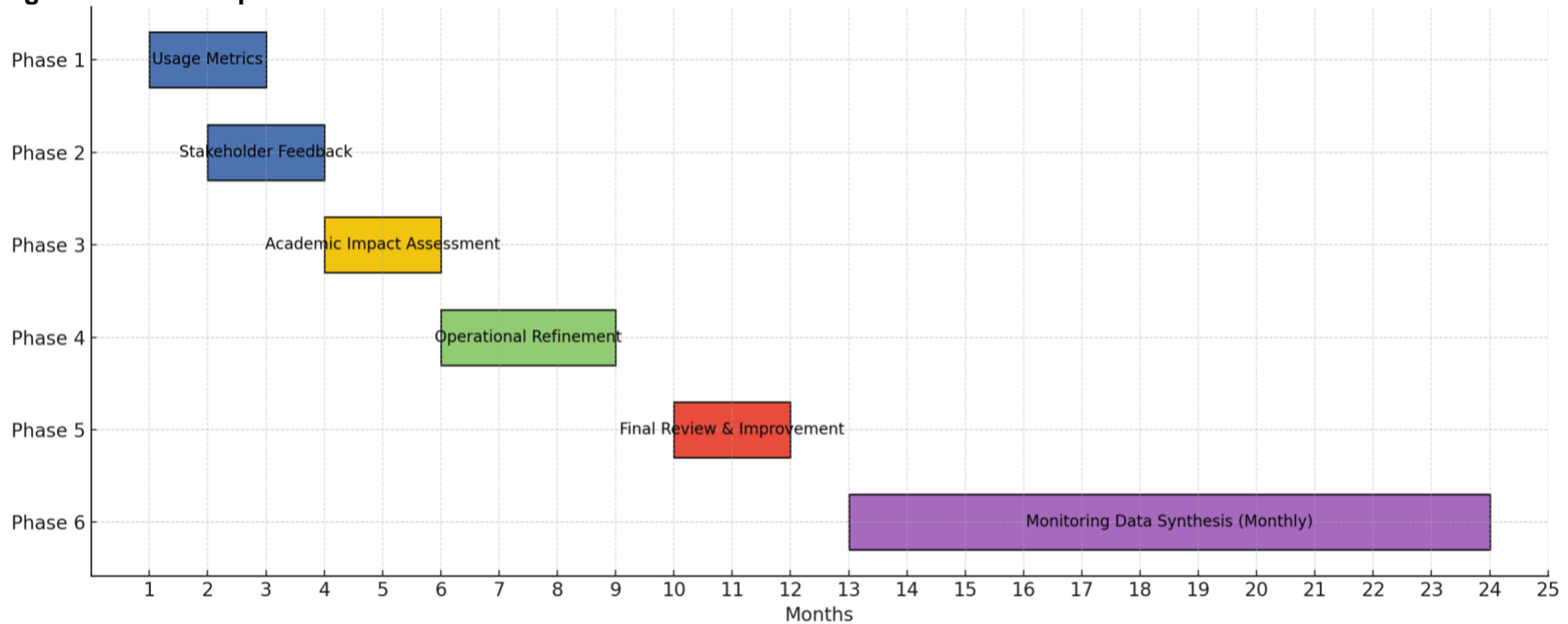
Table A1: Claude Procurement Plan

Step	Action
Assess needs and define requirements	Conduct a formal needs analysis with stakeholders and align outcomes with institutional objectives. Use the needs assessment to define how Claude will be used (e.g., for tutoring and research), ensuring it "most effectively fills the gap" between desired functions and existing tools. De Silva and Alahakoon (2022) state that AI procurement must be grounded in a lifecycle approach from problem conception to deployment, incorporating risk and stakeholder impact from the outset.
Compare options (build vs. buy)	Since Claude is a commercial AI service, the decision is essentially to buy; however, the institution should decide between subscription models or developing supplemental tools. Forum guidance stresses weighing costs and benefits to find the best approach to meet requirements.
Evaluate financing models	Analyze purchasing versus leasing for any needed hardware (servers, specialized workstations). Forum guidance notes that leasing "spreads the cost" of equipment over time, though it may incur higher total payments. The plan should weigh the total cost of ownership against operational budget constraints. For example, a fair-market-value lease could allow biennial upgrades (standard in education) with manageable payments.
Formal RFP and vendor selection	Develop an explicit Request for Proposal (RFP) that specifies technical and legal requirements (security, data privacy, academic use cases). Involve legal, procurement, and IT security offices in drafting contractual terms. EDUCAUSE notes that technology procurement in higher ed is complex and highly collaborative, with many stakeholders (CIO, legal, privacy) guiding decisions. The RFP should require vendors to address how Claude handles institutional data and complies with privacy standards. Consider piloting Claude on a small scale to gather feedback before full deployment.
Negotiate contracts and service levels	Secure favorable terms such as service-level agreements (SLAs) for uptime and support and define failure penalties. The Forum advises that support and maintenance be emphasized: institutions should specify acceptable downtime and remediation commitments in contracts. Avoid selecting a vendor solely on initial cost; ensure long-term support and training via Anthropic are part of the agreement.
Plan for implementation and training	Once a vendor is selected, arrange staff and faculty training on Claude integration leveraging vendor resources (e.g., Anthropic). Assign an IT project manager to coordinate deployment. Follow the Forum's guidance to implement solutions systematically and document decisions and policies throughout the process.

Figure A2: Claude Implementation Milestone Timeline

Appendix B**Table B1: Claude Lifecycle Plan**

Step	Action
Comprehensive asset management	Maintain an up-to-date inventory of all Claude-related devices (laptops, servers, etc.), tagging each with purchase date and user. A detailed inventory provides visibility over procurement timelines and prevents unknown devices from lurking on the network. A structured approach mirrors the concept of a complete "life cycle" framework in AI deployment: De Silva and Alahakoon (2022) describe an inclusive AI lifecycle encompassing all phases from design through deployment.
Performance-based replacement	Use monitoring tools to track device health (CPU/GPU usage, failure rates, battery condition). Data-driven alerts can flag when maintenance costs outweigh value, prompting timely replacement. For example, modern analytics platforms can signal when buying new hardware is cheaper than repairing an old machine. As recommended by CDW and echoed in practitioner guides, refreshing on a schedule (rather than waiting for breakdowns) "smooths out costs" and avoids disruptive failures.
Staggered refresh schedule	Phase replacements (e.g., by department or usage level) so only a portion of devices are refreshed each year, which avoids a massive single-year expense and minimizes downtime. (The Forum's guidance for K–12 suggests staggering by grade; here, we adapt that idea to institutional units.)
Disposal and reuse	Upon retirement, repurpose still-capable devices for low-demand uses (e.g., lab stations, loaner equipment) and recycle end-of-life hardware through certified e-waste programs. Responsible disposal is both ethical and cost-saving. Some universities keep a small "refresh fund" to cover refurbishment or recycling – an approach compatible with federal funding rules like Title III, which allows technology upgrades. Regular refresh cycles support sustainability goals: newer devices are more energy-efficient and environmentally friendly.
Security and sustainability	Remove unsupported hardware from the network when it is retired to close security gaps (old devices often cannot run the latest secure OS). Coordinate refreshes with software updates so all systems can run current, supported software to meet IT security and sustainability objectives. Technology refreshes as a predictable operating expense rather than unpredictable capital bursts, which makes budgeting more reliable. A formal lifecycle plan ensures that Claude's infrastructure remains robust, secure, and aligned with institutional priorities.

Figure B2: Post-Implementation Evaluation Plan Timeline**Phase 1: Usage Metrics (Months 1–3)**

- a. Monitor faculty and student logins, average session time, most-used features, and error reports
- b. Identify adoption rates across departments and highlight early adopters and laggards

Phase 2: Stakeholder Feedback (Months 2–4)

- c. Distribute structured surveys to faculty, students, and IT personnel, assessing:
 - i. Ease of use
 - ii. Helpfulness in teaching or studying
 - iii. Perceived quality of AI responses
 - iv. Conduct 1-on-1 interviews and/or focus groups with 3–5 high-usage departments

Phase 3: Academic Impact Assessment (Months 4–6)

- d. Compare learning analytics (e.g., grades, assignment completion, student engagement) in Claude-supported courses vs. control sections
- e. Track faculty productivity metrics, such as prep time or grading efficiency, if relevant

Phase 4: Operational Refinement and Training Adjustment (Months 6–9)

- f. Use real-world feedback to:
 - i. Identify which implementation components must be sustained, scaled, refined, or replaced. Outcomes will also inform institutional monitoring decisions based on the criteria and replacement triggers outlined in [Table C1](#)
 - ii. Refine internal documentation and training materials
 - iii. Update acceptable use policies and data privacy guidelines
 - iv. Tailor further training sessions based on typical user errors or confusion

Phase 5: Final Review & Continuous Improvement Plan (Months 10–12)

- g. Present findings to leadership (CIO, Provost, Deans).
- h. Adjust the lifecycle replacement and support plan based on identified issues
- i. Incorporate results into broader strategic planning for AI adoption across campus. [Case C2](#) provides a representative case illustrating how usage metrics falling below defined thresholds led to retraining and re-engagement efforts, resulting in measurable improvement.

Phase 6: Monitoring Data Synthesis (Monthly)

- j. Aggregate metrics from [Table C1](#)
- k. Report findings to IT Steering Committee
- l. Trigger refresh protocols when thresholds are breached. See [Chart C3](#) in Appendix C for a decision-mapping visualization of this process and [Case C2](#) for an example where usage dropped below 20 monthly sessions, prompting a retraining initiative that restored engagement by over 60%.

Appendix C**Table C1: Technology Monitoring and Refresh Process**

Category	Monitoring Criteria	Suggested Tools	Replacement Triggers	Sources
AI Tools	User engagement, task success rate, API latency, cost	Built-in dashboards, manual audit logs	< 20 monthly sessions triggers retraining	De Silva et al., 2022; Ooi et al., 2023
Laptops & PCs	CPU load, software compatibility, security patching	Lansweeper, Spiceworks, manual checklists	Battery health < 60%, patch gaps > 60 days	Durand et al., 2019
Network Equipment	Uptime %, throughput, firmware updates	SolarWinds, Open-Audit	MTBF exceeded, firmware EOL, Uptime < 98% or < 500 Mbps avg throughput	Durand et al., 2019
Instructional Tech	Pedagogical alignment, faculty feedback, failure rate	Surveys, usage logs, help desk tickets	3+ tickets/semester OR < 25% faculty adoption	Ooi et al., 2023

Case C2: AI Tool Degradation

Symptoms: Claude session completion rates fell to 63% in the School of Business.

Survey Ratings: Average rating = 2.6/5 from 18 instructors.

Trigger: Engagement < 20 sessions/month and accuracy < 85% for 2+ months.

Action Taken: Pilot retraining with department; adjusted default AI prompts for domain specificity.

Result: Success rate rose to 87% within 6 weeks.

Chart C3: Monthly Monitoring Decision Flowchart